

Finite-deformation theory of olivine carbonation

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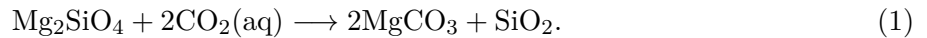
Abstract

We specialize a compositional reacting-mixture theory to an isothermal finite-deformation description of olivine carbonation with three compressible elastic solid phases—forsterite, magnesite and silica—and one aqueous phase. Phase and component mass balances couple mineral conversion to aqueous transport and deformation of a common solid skeleton. A logarithmic elastic free energy satisfies the scalar-distention restriction and determines mineral density, stress and Biot coefficients. Aqueous reactions use mass-based stoichiometric coefficients, while a single fluid free energy determines density and component potentials. A common exchange potential supplies the transfer-work contributions to phase-changing reaction rates and fluid momentum. Pullback to the solid reference configuration gives the balances and weak forms. An automatic-differentiation MOOSE implementation takes time differences of complete reference mass and uses enriched Galerkin diffusion with conservative element fluxes. Actual precipitation and dissolution calculations for a restricted silica-forming reaction are compared with independent reactor and finite-volume reference solutions, including time and mesh refinement. Synthetic checks separately assess constitutive derivatives, transfer work and reference transformations.

1 Physical setting and assumptions

We specialize the component balances, transfer-work normalization and reference measures of the compositional reacting-mixture framework [2]. The elastic and aqueous free energies are application constitutive choices within that framework. The supplement records the equation-by-equation correspondence to the parent. The formulation describes mass transfer, aqueous transport and elastic deformation during carbonation. The solid and fluid constituents occupy a common current region and exchange mass through chemical reactions. Their volume fractions, intrinsic densities and component mass fractions determine local storage. A matched logarithmic elastic energy couples mineral compression to skeleton deformation [3].

The net chemical transformation is



The phase sets are $\mathcal{S} = \{A, B, C\}$, with A forsterite, B magnesite and C silica, and $\mathcal{F} = \{f\}$, the aqueous phase. The index $\xi \in \mathcal{S} \cup \mathcal{F}$ ranges over all phases, and $s \in \mathcal{S}$ ranges over the solids. Each solid is a pure-component phase; aqueous components α are the chemical species enumerated below. All solids share a common skeleton motion. The specialization is isothermal at prescribed $\theta > 0$, elastic, with identity plastic and stress-free factors. A thermal reservoir supplies the heat required to maintain the prescribed temperature.

The aqueous fluid fills the pore volume, so $\phi = \phi_f$ and $S_f = 1$. The interfluid interfacial energy is $\gamma = 0$. Ionic species contribute to charge density, electrochemical transport and the electric field.

We take the permittivity to be constant and common to all phases. The model applies to elastic deformation and reaction within the domain of positive participating phase masses.

2 Phase and component measures and solid kinematics

ρ_ξ^α and ρ_ξ are masses per current mixture volume; $\bar{\rho}_\xi$ is mass per current phase volume. The definitions are

$$\rho_\xi = \phi_\xi \bar{\rho}_\xi = \sum_\alpha \rho_\xi^\alpha, \quad \rho_\xi^\alpha = \phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha, \quad \sum_\alpha \eta_\xi^\alpha = 1, \quad \phi_f + \sum_{s \in \mathcal{S}} \phi_s = 1. \quad (2)$$

For a pure solid its only mass fraction is one. The common skeleton coordinate is $\mathbf{X}$, while $\mathbf{X}_\xi$ denotes a phase-attached reference coordinate when needed. For the mobile fluid, $\mathbf{X}_f$ follows fluid motion, while $\mathbf{X}$ follows the skeleton.

$$\mathbf{x} = \chi(\mathbf{X}, t), \quad \mathbf{F} = \nabla_{\mathbf{X}} \chi, \quad J = \det \mathbf{F} > 0, \quad \mathbf{v}_S = \partial_t \chi|_{\mathbf{X}}, \quad (3)$$

$$\dot{\Gamma} = D_S \Gamma / Dt, \quad D_\xi \Gamma / Dt = \dot{\Gamma} + (\mathbf{v}_\xi - \mathbf{v}_S) \cdot \nabla_{\mathbf{x}} \Gamma. \quad (4)$$

A heavy dot follows the solid skeleton. An ordinary dot on $\mathcal{C}_\xi^\alpha$ follows phase ξ .

The Jacobian J_ξ maps phase-reference volume to current volume; for every solid, $J_s = J$. The phase-reference material increment $\mathcal{C}_\xi^\alpha$, in kg per phase reference volume, contains both conversion and transport relative to the phase:

$$\rho_{\xi 0}^\alpha + \mathcal{C}_\xi^\alpha = J_\xi \phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha, \quad (5)$$

$$\dot{\mathcal{C}}_\xi^\alpha = \frac{\dot{\mathcal{C}}_\xi^\alpha}{J_\xi} = \sum_m \nu_{\xi(m)}^\alpha r_{(m)} - \nabla_{\mathbf{x}} \cdot (\mathbf{j}_{\xi, \text{disp}}^\alpha + \mathbf{j}_{\xi, \text{diff}}^\alpha). \quad (6)$$

Thus $\dot{\mathcal{C}}_\xi^\alpha$ includes reaction production and the divergence of component-relative transport. The separate zero-sum constitutive specialization and the spatial balance are

$$\sum_\alpha \mathbf{j}_{\xi, k}^\alpha = \mathbf{0}, \quad k \in \{\text{disp}, \text{diff}\}, \quad (7)$$

$$\partial_t \rho_\xi^\alpha + \nabla_{\mathbf{x}} \cdot (\rho_\xi^\alpha \mathbf{v}_\xi + \mathbf{j}_{\xi, \text{disp}}^\alpha + \mathbf{j}_{\xi, \text{diff}}^\alpha) = \sum_m \nu_{\xi(m)}^\alpha r_{(m)}. \quad (8)$$

Hence $\sum_\alpha \dot{\mathcal{C}}_f^\alpha$ is the net aqueous phase mass-production rate. For each pure solid, the component-relative fluxes are zero.

The compressible scalar-distention specialization is

$$\mathbf{F} = \mathbf{A}_s \bar{\mathbf{F}}_s, \quad \mathbf{A}_s = a_s^{1/3} \mathbf{I}, \quad J = a_s \bar{J}_s, \quad \bar{J}_s = \det \bar{\mathbf{F}}_s = \frac{\bar{\rho}_{s0}}{\bar{\rho}_s}. \quad (9)$$

Summing the solid component measures gives

$$J \phi_s \bar{\rho}_s = \phi_{s0} \bar{\rho}_{s0} + \mathcal{C}_s, \quad a_s \phi_s = \phi_{s0} + \mathcal{C}_s / \bar{\rho}_{s0}, \quad (10)$$

$$\frac{\dot{a}_s}{a_s} = \nabla_{\mathbf{x}} \cdot \mathbf{v}_S + \frac{\dot{\bar{\rho}}_s}{\bar{\rho}_s}, \quad \dot{\mathcal{C}}_s = J \sum_m \nu_{s(m)} r_{(m)}. \quad (11)$$

Taking determinants of the deformation split gives $a_s = J / \bar{J}_s$, including during mass exchange. The adopted true-material density relation $\bar{J}_s \bar{\rho}_s = \bar{\rho}_{s0}$, together with phase mass balance, gives $a_s \phi_s = \phi_{s0} + \mathcal{C}_s / \bar{\rho}_{s0}$. Thus the accumulated mass exchange enters the relation between distention and phase volume fraction through $\mathcal{C}_s$. When $\mathcal{C}_s = 0$, this relation reduces to $a_s = \phi_{s0} / \phi_s$. Reaction can change ϕ_s at fixed J through the material increment. The constitutive equations and boundary conditions determine the resulting stress.

3 Variational statement and transfer work

At a fixed instantaneous thermal state, coordinate stationarity gives

$$\delta \int_{t_1}^{t_2} (T - U) dt + \int_{t_1}^{t_2} \left(\delta W + \sum_{k=1}^4 \delta C_k \right) dt = 0. \quad (12)$$

Let $\mathcal{B}$ denote the current mixture region and $\mathcal{B}_{\xi 0}$ the phase-reference region. The body acceleration is $\mathbf{g}$, the interaction force per mixture volume is $\mathbf{b}_\xi$, and $\mathbf{t}_\xi$ is phase traction per current boundary area. The electrical variables φ , ϱ , and $\bar{q}$ denote potential, charge per mixture volume, and outward electric displacement. For these four phases, the energies and external virtual work are

$$T = \sum_{\xi} \int_{\mathcal{B}} \frac{1}{2} \rho_{\xi} |\mathbf{v}_{\xi}|^2 dv, \quad U = \sum_{\xi} \int_{\mathcal{B}} (\rho_{\xi} \psi_{\xi} + \phi_{\xi} \omega_{\xi}^+) dv, \quad (13)$$

$$\begin{aligned} \delta W = \int_{\mathcal{B}} \left[\sum_{\xi} (\rho_{\xi} \mathbf{g} + \mathbf{b}_{\xi}) \cdot \delta \mathbf{x}_{\xi} + \sum_{\xi, \alpha} (\psi_{\xi} + L_{\xi}^{\alpha}) \frac{\delta \mathcal{C}_{\xi}^{\alpha}}{J_{\xi}} - \varrho \delta \varphi \right] dv \\ + \sum_{\xi} \int_{\partial \mathcal{B}} \mathbf{t}_{\xi} \cdot \delta \mathbf{x}_{\xi} da + \int_{\partial \mathcal{B}} \bar{q} \delta \varphi da. \end{aligned} \quad (14)$$

The four constraints are

$$\begin{aligned} C_1 &= \int_{\mathcal{B}} \lambda (\sum_{\xi} \phi_{\xi} - 1) dv, \\ C_2 &= \sum_{\xi, \alpha} \int_{\mathcal{B}_{\xi 0}} \pi_{\xi}^{\alpha} \left(J_{\xi} - \frac{\rho_{\xi 0}^{\alpha} + \mathcal{C}_{\xi}^{\alpha}}{\phi_{\xi} \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}} \right) dV_{\xi}, \\ C_3 &= \sum_{\xi} \int_{\mathcal{B}} \Pi_{\xi} (\sum_{\alpha} \eta_{\xi}^{\alpha} - 1) dv, \\ C_4 &= \int_{\mathcal{B}} \tau \sum_{\xi, \alpha} \frac{\dot{\mathcal{C}}_{\xi}^{\alpha}}{J_{\xi}} dv. \end{aligned} \quad (15)$$

The component sums in a pure solid have one term. The ratios in these constraints are evaluated for positive component masses; phase boundaries require an active-phase limit.

The material-measure variation gives $\delta \int \rho_{\xi}^{\alpha} \Upsilon_{\xi} dv = \int [\rho_{\xi}^{\alpha} \delta \Upsilon_{\xi} + \Upsilon_{\xi} \delta \mathcal{C}_{\xi}^{\alpha} / J_{\xi}] dv$. Thus the inserted Helmholtz storage in $-\delta U$ cancels the $+\psi_{\xi}$ supply in δW . Time integration of kinetic work gives both $-\rho_{\xi} \mathbf{a}_{\xi}$ and $-\sum_{\alpha} \dot{\mathcal{C}}_{\xi}^{\alpha} \mathbf{v}_{\xi}$. The material-insertion constraint gives the complementary gradient term. Combining these variations gives the transfer relation

$$\frac{D_{\xi} \tau}{Dt} = \frac{1}{2} |\mathbf{v}_{\xi}|^2 + L_{\xi}^{\alpha} - \frac{\pi_{\xi}^{\alpha}}{\phi_{\xi} \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}}, \quad (16)$$

$$\hat{\mu}_{\xi}^{\alpha} = \psi_{\xi} + \frac{\pi_{\xi}^{\alpha}}{\phi_{\xi} \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}}, \quad \mu_{\xi}^{\alpha} = \hat{\mu}_{\xi}^{\alpha} + z_{\xi}^{\alpha} \varphi. \quad (17)$$

Here $\hat{\mu}$, μ , L , and ψ have units J/kg. The specific charge z_{ξ}^{α} has units C/kg.

Choose any present solid $s_{\star}$ as the transfer reference. The normalization and its immediate consequence are

$$\psi_{s_{\star}} + L_{s_{\star}}^N = \hat{\mu}_{s_{\star}}^N, \quad \dot{\tau} = \frac{1}{2} |\mathbf{v}_S|^2, \quad (18)$$

$$L_{\xi}^{\alpha} = \hat{\mu}_{\xi}^{\alpha} - \psi_{\xi} + D_{\xi} \tau / Dt - \frac{1}{2} |\mathbf{v}_{\xi}|^2. \quad (19)$$

The single exchange potential τ has a prescribed initial field and a fixed time-independent additive constant. For a stationary skeleton, (18) preserves that initial field. Any present solid can serve as the transfer reference because all solids share the same velocity and normalization.

Writing $\mathbf{v}_f = \mathbf{v}_S + \mathbf{w}_f/\rho_f$ gives the explicit fluid transfer offset:

$$\frac{D_s \tau}{Dt} - \frac{1}{2} |\mathbf{v}_S|^2 = 0, \quad \frac{D_f \tau}{Dt} - \frac{1}{2} |\mathbf{v}_f|^2 = \frac{\mathbf{w}_f}{\rho_f} \cdot (\nabla_{\mathbf{x}} \tau - \mathbf{v}_S) - \frac{1}{2} \left| \frac{\mathbf{w}_f}{\rho_f} \right|^2. \quad (20)$$

The offset is retained in phase-changing reaction power below.

4 Elastic free energy and constitutive restrictions

With all plastic and stress-free factors equal to identity, we use the constitutive state $\psi_s(\bar{\mathbf{F}}_s, \mathbf{A}_s, \bar{\rho}_s, \theta)$. For isotropic distention its constitutive restrictions include

$$\bar{p}_s = \bar{\rho}_s^2 \frac{\partial \psi_s}{\partial \bar{\rho}_s} \Big|_{\bar{\mathbf{F}}_s, a_s}, \quad \boldsymbol{\sigma}'_s = \rho_s \frac{\partial \psi_s}{\partial \bar{\mathbf{F}}_s} \Big|_{a_s, \bar{\rho}_s} \bar{\mathbf{F}}_s^T, \quad \frac{1}{3} \text{tr} \boldsymbol{\sigma}'_s = \rho_s a_s \frac{\partial \psi_s}{\partial a_s} \Big|_{\bar{\mathbf{F}}_s, \bar{\rho}_s}. \quad (21)$$

The electrical enthalpy selected below has zero density derivative. The constitutive derivatives treat kinematics and density as independent arguments before imposing their mass-conservation relation: $J = a_s \det \bar{\mathbf{F}}_s$ is a kinematic argument, while $\bar{\rho}_{s0}/\bar{\rho}_s$ is a density argument. Their relation to $\bar{J}_s$ is imposed *after* taking the indicated partial derivatives.

Let $K_s > 0$ be the reference intrinsic mineral modulus, $G_s > 0$ a phase-normalized skeleton shear modulus, and $0 < K < \phi_{s0} K_s$ the drained skeleton bulk modulus allocated to solid s per unit mixture reference volume. We use the same symbol K for each solid branch; its value may differ by mineral, with the branch understood from s . These material parameters specify the elastic response of each branch. The reference fraction ϕ_{s0} converts K to the phase-normalized coefficient K/ϕ_{s0} . An admissible energy is

$$\psi_s = \psi_s^0 + \frac{1}{\bar{\rho}_{s0}} \left\{ \frac{G_s}{2} [J^{-2/3} \mathbf{F} : \mathbf{F} - 3] + \frac{K}{2\phi_{s0}(1 - K/(\phi_{s0}K_s))} \left[\ln \frac{J}{\bar{\rho}_{s0}/\bar{\rho}_s} \right]^2 + \frac{K_s}{2} \left[\ln \frac{\bar{\rho}_{s0}}{\bar{\rho}_s} \right]^2 \right\}. \quad (22)$$

ψ_s^0 is the chemically consistent reference Helmholtz energy per mass at the prescribed temperature. It contributes to reaction potentials, and its mechanical derivatives are zero. The mixture reference storage is $\sum_s J \rho_s \psi_s$, with the reacting mass $J \rho_s = \phi_{s0} \bar{\rho}_{s0} + \mathcal{C}_s$. Added material takes the recipient phase state, under local equilibrium within each phase. Phase connectivity and the modulus allocation in (22) are assumptions requiring calibration.

Differentiation before enforcing true mass conservation, then substitution of $\bar{\rho}_{s0}/\bar{\rho}_s = \bar{J}_s$, gives

$$\bar{p}_s = \frac{[K/(\phi_{s0}(1 - K/(\phi_{s0}K_s)))] \ln(J/\bar{J}_s) - K_s \ln \bar{J}_s}{\bar{J}_s}, \quad (23)$$

$$\boldsymbol{\sigma}'_s = \frac{\rho_s}{\bar{\rho}_{s0}} \left[G_s J^{-2/3} \text{dev}(\mathbf{F}\mathbf{F}^T) + \frac{K \ln(J/\bar{J}_s)}{\phi_{s0}(1 - K/(\phi_{s0}K_s))} \mathbf{I} \right]. \quad (24)$$

Because the isochoric term is invariant under scalar changes in a_s , both the true-deformation trace and the distention derivative in (21) equal $\rho_s K \ln(J/\bar{J}_s)/[\bar{\rho}_{s0}\phi_{s0}(1 - K/(\phi_{s0}K_s))]$. Thus the energy satisfies the scalar-distention restriction in (21).

The intrinsic mechanical mineral mean stress is recovered from the same pair of derivatives:

$$\frac{1}{3\phi_s} \text{tr} \boldsymbol{\sigma}'_s - \bar{p}_s = \frac{K_s \ln \bar{J}_s}{\bar{J}_s}. \quad (25)$$

This expression gives the intrinsic mechanical mean stress. The electric field contributes the Maxwell stress defined below.

4.1 Electrical state, pressure identification and scalar mineral solve

For definiteness choose a common constant permittivity $\epsilon > 0$ and a linear electric enthalpy. The exact specialized electrical and volume-fraction relations are

$$\mathbf{E} = -\nabla_{\mathbf{x}}\varphi, \quad \omega_{\xi}^+ = -\frac{1}{2}\epsilon|\mathbf{E}|^2, \quad \mathbf{d}_{\xi} = \epsilon\mathbf{E}, \quad \mathbf{d} = \sum_{\xi} \phi_{\xi}\mathbf{d}_{\xi} = \epsilon\mathbf{E}, \quad (26)$$

$$\bar{p}_{\xi} - \omega_{\xi}^+ = \bar{p}_E = -\lambda, \quad \bar{p}_s = \bar{p}_f, \quad \nabla_{\mathbf{x}} \cdot \mathbf{d} = \varrho = \sum_{\alpha} z_f^{\alpha} \phi_f \bar{p}_f \eta_f^{\alpha}. \quad (27)$$

In an electric field, $\bar{p}_E = \bar{p}_f + \epsilon|\mathbf{E}|^2/2$. For $\mathbf{E} = \mathbf{0}$ both reduce to the common pore pressure. The pure solids are uncharged. Gauss' law resolves the field associated with the ionic charge density, including formal Debye-scale effects.

Using $\bar{p}_s = \bar{p}_f$, the local mineral equation is

$$K_s \ln \bar{J}_s + \left(1 - \frac{K}{\phi_{s0}K_s}\right) \bar{p}_f \bar{J}_s - \frac{K}{\phi_{s0}} \ln J = 0. \quad (28)$$

It has a unique positive root at $\bar{p}_f \geq 0$. For negative pressure, use the branch continuous from zero pressure only while $K_s + (1 - K/(\phi_{s0}K_s))\bar{p}_f \bar{J}_s > 0$. Also require $0 < \bar{J}_s < e$ for positive tangent of the prescribed mineral mean-stress law, positive phase masses and positive pore volume. Constitutive reference constants and chemical standard states must use the same pressure origin; $\bar{p}_f = 0$ defines the mechanical pressure origin.

At fixed $\bar{p}_E$, electric field and instantaneous reacted mass per reference volume, implicit differentiation gives

$$\left. \frac{\partial \bar{J}_s}{\partial J} \right|_{\bar{p}_E, \mathbf{E}} = \frac{K \bar{J}_s}{\phi_{s0}J[K_s + (1 - K/(\phi_{s0}K_s))\bar{p}_f \bar{J}_s]}, \quad 1 - \bar{B}_s = a_s \left. \frac{\partial \bar{J}_s}{\partial J} \right|_{\bar{p}_E, \mathbf{E}}. \quad (29)$$

The last equality follows directly from the specific-volume definition with $\bar{v}_s = \bar{J}_s/\bar{\rho}_{s0}$.

$$\boldsymbol{\sigma}_s'' = \boldsymbol{\sigma}_s' - \phi_s(1 - \bar{B}_s)\bar{p}_E \mathbf{I}, \quad \boldsymbol{\sigma}'' = \sum_s \boldsymbol{\sigma}_s'', \quad (30)$$

$$B = 1 - \sum_s \phi_s(1 - \bar{B}_s), \quad \mathbf{P}'' = J \boldsymbol{\sigma}'' \mathbf{F}^{-T}, \quad (31)$$

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}'' - B \bar{p}_E \mathbf{I} + \mathbf{E} \otimes \mathbf{d} = \sum_s \boldsymbol{\sigma}_s' - \bar{p}_f \mathbf{I} + \epsilon(\mathbf{E} \otimes \mathbf{E} - \frac{1}{2}|\mathbf{E}|^2 \mathbf{I}). \quad (32)$$

The final equality is an algebraic check on electrical pressure bookkeeping for common permittivity. The double-prime stress is evaluated at the current prescribed pressure; the drained response follows by setting pressure and electric field to zero.

4.2 Elastic limiting cases

For a single nonreacting solid, $J\rho_s = \phi_{s0}\bar{\rho}_{s0}$. In the matched logarithmic elastic law, identify the bulk moduli with K and K_s , and the shear modulus with $G = \phi_{s0}G_s$. Then $J\rho_s(\psi_s - \psi_s^0)$ equals its coupled elastic energy, and (28) is exactly its elastic mineral residual. The mineral modulus remains K_s .

At zero pressure and zero field, $\bar{J}_s = J^{K/(\phi_{s0}K_s)}$. The drained mixture reference energy is $\sum_s(\phi_{s0} + \mathcal{C}_s/\bar{\rho}_{s0})[G_s(J^{-2/3}\mathbf{F} : \mathbf{F} - 3)/2 + (K/\phi_{s0})(\ln J)^2/2]$. For a frozen reacted state about $J = 1$,

$$B_0 = 1 - \sum_s \frac{\phi_s K}{\phi_{s0} K_s}, \quad K_{\text{drained}} = \sum_s \frac{\phi_s K}{\phi_{s0}}, \quad G_{\text{drained}} = \sum_s \phi_s G_s. \quad (33)$$

These are tangent limits at the stated state. At the unreacted reference state, each participating solid contributes its allocated K to the drained modulus. In the incompressible-mineral limit $K_s \rightarrow \infty$, $\bar{J}_s \rightarrow 1$, $a_s \rightarrow J$, and $B \rightarrow 1$; these are the incompressible-mineral kinematics.

5 Aqueous components and chemistry

The aqueous components are the species in Table 1. All dissolved species belong to the aqueous phase. Here M_α is species molar mass, F_{Far} is Faraday’s constant, and the integer in the last column is the valence used to compute the specific charge z_f^α . Atomic masses are used consistently; electron mass is neglected in the chemical mass bookkeeping.

α	Species	Mg	Si	C	H	O	$M_\alpha z_f^\alpha / F_{\text{Far}}$
1	H ⁺	0	0	0	1	0	1
2	OH [−]	0	0	0	1	1	-1
3	CO ₂ (aq)	0	0	1	0	2	0
4	HCO ₃ [−]	0	0	1	1	3	-1
5	CO ₃ ^{2−}	0	0	1	0	3	-2
6	Mg ²⁺	1	0	0	0	0	2
7	H ₄ SiO ₄	0	1	0	4	4	0
8	H ₃ SiO ₄ [−]	0	1	0	3	4	-1
$N = 9$	H ₂ O	0	0	0	2	1	0

Table 1: A minimal application species set within the single aqueous phase. Other aqueous complexes require additional components and compatible free-energy data.

5.1 One free energy for density and component potentials

We use $\psi_f(\bar{\rho}_f, \{\eta_f^\alpha\}, \theta)$ and $\bar{p}_f = \bar{\rho}_f^2 \partial \psi_f / \partial \bar{\rho}_f$. An ideal-mixture example can be specified through its mass-specific Gibbs transform, with mole fractions used only as constitutive functions of the mass fractions:

$$x_f^\alpha = \frac{\eta_f^\alpha / M_\alpha}{\sum_\beta \eta_f^\beta / M_\beta}, \quad \bar{v}_f^0 = \sum_\alpha \frac{\eta_f^\alpha v_\alpha^0}{M_\alpha}, \quad (34)$$

$$g_f(\bar{p}_f, \{\eta_f^\alpha\}) = \sum_\alpha \frac{\eta_f^\alpha}{M_\alpha} \left[g_\alpha^0 + R\theta \ln x_f^\alpha + v_\alpha^0 K_f (1 - e^{-(\bar{p}_f - p_0)/K_f}) \right]. \quad (35)$$

g_α^0 is a standard molar chemical potential, $v_\alpha^0 > 0$ a constant molar volume and $K_f > 0$ a pressure modulus. These constitutive parameters require material calibration. p_0 uses the same pressure origin as $\bar{p}_f$. This example assumes ideal mixing. A nonideal model may replace it only by supplying a differentiable thermodynamic potential and its consistent derivatives.

At fixed mass fractions the Legendre derivative and inverse give

$$\frac{1}{\bar{\rho}_f} = \left. \frac{\partial g_f}{\partial \bar{p}_f} \right|_{\eta} = \bar{v}_f^0 e^{-(\bar{p}_f - p_0)/K_f}, \quad \bar{p}_f = p_0 + K_f \ln(\bar{\rho}_f \bar{v}_f^0), \quad (36)$$

$$\psi_f(\bar{\rho}_f, \eta) = g_f(\bar{p}_f(\bar{\rho}_f, \eta), \eta) - \frac{\bar{p}_f(\bar{\rho}_f, \eta)}{\bar{\rho}_f}, \quad (37)$$

$$\hat{\mu}_f^\alpha = \frac{1}{M_\alpha} [g_\alpha^0 + R\theta \ln x_f^\alpha + v_\alpha^0 K_f (1 - e^{-(\bar{p}_f - p_0)/K_f})]. \quad (38)$$

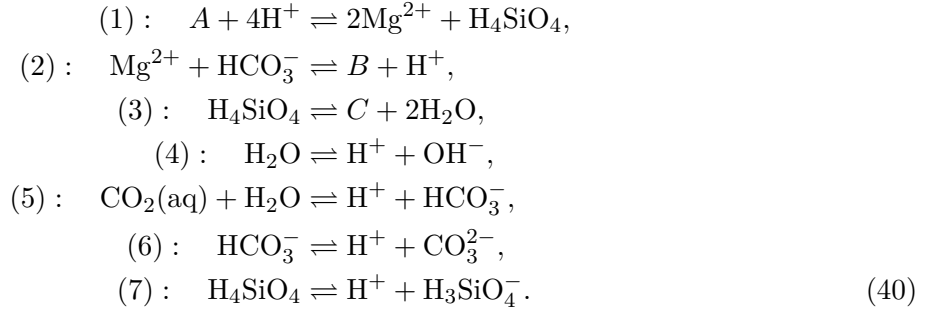
For example, $\partial\psi_f/\partial\bar{\rho}_f$ contains the chain terms $(\partial g_f/\partial\bar{p}_f - 1/\bar{\rho}_f)\partial\bar{p}_f/\partial\bar{\rho}_f$, which cancel, leaving $\bar{p}_f/\bar{\rho}_f^2$. The constituent-mass derivative, taken at fixed phase volume and other component densities, yields (38). Constant density-independent electrical enthalpy adds zero to this derivative. The pure-solid counterpart is

$$\hat{\mu}_s = \psi_s + \bar{p}_s/\bar{\rho}_s, \quad s \in \{A, B, C\}. \quad (39)$$

The reference chemical energies in ψ_s^0 and g_α^0 must share an elemental reference convention. Compatible chemical data determine reaction direction and equilibrium composition.

5.2 Mass-based reaction mechanisms

The three mineral mechanisms and four aqueous mechanisms are chosen as



Here $r_{(m)}$ is chosen to be molar progress per *current mixture* volume and time, $\text{mol m}^{-3}\text{s}^{-1}$. Each $\nu_{\xi(m)}^\alpha$ is its signed molar stoichiometric number multiplied by M_α , in kg/mol. For example, $\nu_{A(1)} = -M_A$, $\nu_{f(1)}^1 = -4M_1$, $\nu_{f(1)}^6 = 2M_6$, $\nu_{f(1)}^7 = M_7$. Reference-volume production is $J\nu_{\xi(m)}^\alpha r_{(m)}$.

Every column satisfies the mass and charge constraints, and atom conservation gives further exact checks:

$$\sum_{\xi, \alpha} \nu_{\xi(m)}^\alpha = 0, \quad \sum_{\xi, \alpha} z_\xi^\alpha \nu_{\xi(m)}^\alpha = 0, \quad \sum_{\xi, \alpha} \frac{n_e^{\xi\alpha}}{M_{\xi\alpha}} \nu_{\xi(m)}^\alpha = 0. \quad (41)$$

$n_e^{\xi\alpha}$ is the number of atoms of element e in one formula unit. Element balances are obtained by weighting the component mass balances with n_e/M . Mechanisms (1), twice (2), (3), and twice (5) sum to (1). Independent rates allow dissolved Mg and Si to accumulate or travel before precipitation. The local rate ratios follow from the individual kinetic laws.

6 Reaction and transport laws

At common temperature the uncoupled Onsager reaction law becomes

$$r_{(m)} = -\frac{K_{(m)}}{\theta} \sum_{\xi, \alpha} (\psi_\xi + L_\xi^\alpha) \nu_{\xi(m)}^\alpha, \quad K_{(m)} > 0, \quad (42)$$

$$\mathcal{A}_{(m)} = -\sum_{\xi, \alpha} \mu_\xi^\alpha \nu_{\xi(m)}^\alpha = -\sum_{\xi, \alpha} \hat{\mu}_\xi^\alpha \nu_{\xi(m)}^\alpha, \quad (43)$$

$$-\sum_{\xi, \alpha} (\psi_\xi + L_\xi^\alpha) \nu_{\xi(m)}^\alpha = \mathcal{A}_{(m)} - \left(D_f \tau / Dt - \frac{1}{2} |\mathbf{v}_f|^2 \right) \sum_{\alpha} \nu_{f(m)}^\alpha. \quad (44)$$

The electrical potential cancels by charge conservation, while the solid transfer offsets vanish by (18). The fluid offset contributes to dissolution and precipitation through the net aqueous mass transfer in (44). $K_{(m)}$ has units $\text{mol}^2\text{K}/(\text{J m}^3\text{s})$ under the declared molar progress convention and is positive in the interior of the admissible composition domain. These smooth rate laws apply to positive participating masses. Phase disappearance and nucleation require an active-set or degenerate-mobility extension.

For homogeneous mechanisms (4)–(7), $\sum_{\alpha} \nu_{f(m)}^\alpha = 0$. Their rapid-equilibrium limit is consequently $\mathcal{A}_{(m)} = 0$, and (38) gives

$$\prod_{\alpha} (x_f^\alpha)^{\nu_{f(m)}^\alpha / M_\alpha} = \exp \left[-\frac{1}{R\theta} \sum_{\alpha} \frac{\nu_{f(m)}^\alpha}{M_\alpha} \{g_\alpha^0 + v_\alpha^0 K_f (1 - e^{-(\bar{p}_f - p_0)/K_f})\} \right], \quad m = 4, \dots, 7. \quad (45)$$

Equation (45) is the usual mass-action equilibrium relation [1], specialized to the ideal-mixture chemical potentials in (38); its pressure dependence follows from the chosen Gibbs energy in (35). The finite-rate system uses all seven laws in (42). In the optional rapid-equilibrium limit, (45) replaces the four aqueous rate laws; their four rates then act as algebraic multipliers determined by the species balances.

The isothermal transport law, with water as reference component N , is

$$\begin{aligned} \mathbf{j}_{f,k}^\alpha &= -\sum_{\beta \neq N} \mathbf{D}_{f,k}^{\alpha\beta} [\nabla_{\mathbf{x}} (\hat{\mu}_f^\beta - \hat{\mu}_f^N) + (z_f^\beta - z_f^N) \nabla_{\mathbf{x}} \varphi], \quad \alpha \neq N, \\ \mathbf{j}_{f,k}^N &= -\sum_{\alpha \neq N} \mathbf{j}_{f,k}^\alpha, \quad k \in \{\text{diff}, \text{disp}\}. \end{aligned} \quad (46)$$

Each reduced mobility is symmetric positive definite on its eight independent mass-flux coordinates, with units $\text{kg s}/\text{m}^3$. Mechanical dispersion can be omitted by setting that entire constitutive block to zero and omitting its inverse in dissipation. Different mobilities or nonideal chemistry remain constitutive choices for these forces. A component-independent phase potential cancels from the gradient differences; spatially constant electrical gauge shifts leave the driving gradients unchanged.

Weighting the component balances by specific charge and summing gives the charge balance:

$$\mathbf{i} = \sum_{\alpha} z_f^\alpha (\rho_f^\alpha \mathbf{v}_f + \mathbf{j}_{f,\text{disp}}^\alpha + \mathbf{j}_{f,\text{diff}}^\alpha), \quad \partial_t \varrho + \nabla_{\mathbf{x}} \cdot \mathbf{i} = 0. \quad (47)$$

Gauss' law in (27) supplies the field equation. Pure electric-displacement boundary data require the compatibility condition $\int_{\mathcal{B}} \varrho \, dv = \int_{\partial \mathcal{B}} \bar{q} \, da$ and its rate; a potential boundary allows its surface charge to adjust. Prescribed electric and ionic fluxes must satisfy this compatibility condition throughout the evolution.

7 Fluid momentum, the reacting flux and overall momentum

Let $\mathbf{a}_\xi = D_\xi \mathbf{v}_\xi / Dt$ denote phase acceleration. With $S_f = 1$ and $\gamma = 0$, fluid momentum balance includes electrical and material-insertion forces:

$$\rho_f \mathbf{a}_f = \rho_f \mathbf{g} + \mathbf{b}_f - \phi_f \nabla_{\mathbf{x}} \bar{p}_E + \nabla_{\mathbf{x}} \cdot (\phi_f \mathbf{E} \otimes \mathbf{d}_f) + \sum_{\alpha} \dot{c}_f^{\alpha} (\nabla_{\mathbf{x}} \tau - \mathbf{v}_f). \quad (48)$$

Viscous drag opposes fluid motion relative to the skeleton. The relative mass flux and its drag closure are

$$\mathbf{w}_f = \rho_f (\mathbf{v}_f - \mathbf{v}_S), \quad \mathbf{b}_f = -\frac{\mu_f^{\text{visc}} \phi_f}{\bar{\rho}_f} \boldsymbol{\kappa}_f^{-1} \mathbf{w}_f, \quad \boldsymbol{\kappa}_f = \boldsymbol{\kappa}_f^T > 0. \quad (49)$$

Here the viscosity is the aqueous dynamic viscosity, and the permeability is the symmetric positive-definite tensor in the drag law. Substitution of $\mathbf{v}_f = \mathbf{v}_S + \mathbf{w}_f / \rho_f$ moves the inserted-mass velocity term to the left:

$$\begin{aligned} \frac{1}{\rho_f} \left(\phi_f^2 \mu_f^{\text{visc}} \boldsymbol{\kappa}_f^{-1} + \sum_{\alpha} \dot{c}_f^{\alpha} \mathbf{I} \right) \mathbf{w}_f &= \rho_f (\mathbf{g} - \mathbf{a}_f) - \phi_f \nabla_{\mathbf{x}} \bar{p}_E \\ &\quad + \nabla_{\mathbf{x}} \cdot (\phi_f \mathbf{E} \otimes \mathbf{d}_f) + \sum_{\alpha} \dot{c}_f^{\alpha} (\nabla_{\mathbf{x}} \tau - \mathbf{v}_S), \end{aligned} \quad (50)$$

$$\tilde{\boldsymbol{\lambda}}_f = \left(\phi_f^2 \mu_f^{\text{visc}} \boldsymbol{\kappa}_f^{-1} + \sum_{\alpha} \dot{c}_f^{\alpha} \mathbf{I} \right)^{-1}. \quad (51)$$

Thus $\mathbf{w}_f$ is $\rho_f \tilde{\boldsymbol{\lambda}}_f$ times the entire right-hand side of (50). The resistance has units $\text{kg m}^{-3} \text{s}^{-1}$, $\nabla \tau$ has velocity units, and each right-hand term has force-per-volume units. The conversion term in the resistance and the $\nabla \tau$ force are the two parts of the same phase momentum derivation. Both contributions enter the coupled reaction and flow problem.

Algebraic inversion requires every shifted resistance eigenvalue to be nonzero. Using the largest permeability eigenvalue, a stronger positive-resistance regime is

$$\sum_{\alpha} \dot{c}_f^{\alpha} > -\frac{\phi_f^2 \mu_f^{\text{visc}}}{\kappa_{f,\max}}. \quad (52)$$

The conversion shift enters the momentum resistance, while the drag law supplies the dissipative power. For negligible fluid inertia one sets $\mathbf{a}_f = \mathbf{0}$ *after* the derivation. With zero field and zero phase conversion, the Darcy limit is

$$\phi_f (\mathbf{v}_f - \mathbf{v}_S) = -\frac{\boldsymbol{\kappa}_f}{\mu_f^{\text{visc}}} (\nabla_{\mathbf{x}} \bar{p}_f - \bar{\rho}_f \mathbf{g}). \quad (53)$$

Summing the phase momenta uses $\sum_{\xi} \mathbf{b}_{\xi} = 0$ and $\sum_{\xi, \alpha} \dot{c}_{\xi}^{\alpha} = 0$. All solids move with $\mathbf{v}_S$, so the global $\nabla \tau$ term cancels and insertion momentum takes the form

$$\sum_{\xi} \rho_{\xi} \mathbf{a}_{\xi} = \nabla_{\mathbf{x}} \cdot \boldsymbol{\sigma} + \rho \mathbf{g} - \left(\sum_{\alpha} \dot{c}_f^{\alpha} \right) (\mathbf{v}_f - \mathbf{v}_S), \quad \rho = \sum_{\xi} \rho_{\xi}. \quad (54)$$

The phase-relative transfer contribution remains in (48) and (44).

8 Thermodynamic restriction

The mechanical, pressure, composition and volume-fraction restrictions above remove the reversible rate coefficients in the Coleman–Noll inequality. Uniform prescribed temperature removes temperature-gradient terms; elasticity removes plastic dissipation; one pore phase removes saturation and interfacial-history dissipation. The remaining entropy production is

$$\begin{aligned}
0 \leq & -\frac{1}{\theta} \sum_m r_{(m)} \sum_{\xi, \alpha} (\psi_\xi + L_\xi^\alpha) \nu_{\xi(m)}^\alpha \\
& - \frac{1}{\theta} \sum_{k, \alpha \neq N} \mathbf{j}_{f,k}^\alpha \cdot [\nabla_{\mathbf{x}} (\hat{\mu}_f^\alpha - \hat{\mu}_f^N) + (z_f^\alpha - z_f^N) \nabla_{\mathbf{x}} \varphi] - \frac{1}{\theta} \mathbf{b}_f \cdot (\mathbf{v}_f - \mathbf{v}_S) \\
= & \sum_m \frac{r_{(m)}^2}{K_{(m)}} + \frac{1}{\theta} \sum_{k, \alpha, \beta \neq N} \mathbf{j}_{f,k}^\alpha \cdot (\mathbf{D}_{f,k}^{-1})^{\alpha\beta} \mathbf{j}_{f,k}^\beta + \frac{\phi_f^2 \mu_f^{\text{visc}}}{\theta} (\mathbf{v}_f - \mathbf{v}_S) \cdot \boldsymbol{\kappa}_f^{-1} (\mathbf{v}_f - \mathbf{v}_S). \quad (55)
\end{aligned}$$

The mechanical interaction-energy allocation places half the nonnegative drag heating in the aqueous subsystem and half in the aggregate solid subsystem. Their sum cancels the loss of relative mechanical power and conserves mixture energy. At a common temperature any additional equal-and-opposite reaction-heat allocation cancels from the total entropy supply. The thermostat removes the total heat required by the first law. The transfer force in (55) is given by (44).

9 Solid-reference balances and weak forms

Scalar gradients and fluxes pull back by

$$\nabla_{\mathbf{x}} = \mathbf{F}^{-T} \nabla_{\mathbf{X}}, \quad \mathbf{W}_f = J \mathbf{F}^{-1} \mathbf{w}_f, \quad \tilde{\Lambda}_f = J \mathbf{F}^{-1} \tilde{\lambda}_f \mathbf{F}^{-T}, \quad J \nabla_{\mathbf{x}} \cdot \mathbf{w}_f = \nabla_{\mathbf{X}} \cdot \mathbf{W}_f. \quad (56)$$

The reference flux relation, including the electrical force, is

$$\begin{aligned}
\mathbf{W}_f = \rho_f \tilde{\Lambda}_f \Big[& \rho_f \mathbf{F}^T (\mathbf{g} - \mathbf{a}_f) - \phi_f \nabla_{\mathbf{x}} \bar{p}_E + \mathbf{F}^T \nabla_{\mathbf{x}} \cdot (\phi_f \mathbf{E} \otimes \mathbf{d}_f) \\
& + \sum_{\alpha} \dot{c}_f^\alpha (\nabla_{\mathbf{x}} \tau - \mathbf{F}^T \mathbf{v}_S) \Big]. \quad (57)
\end{aligned}$$

The component balances in the solid reference configuration are

$$\partial_t (J \phi_f \bar{\rho}_f \eta_f^\alpha) |_{\mathbf{X}} + \nabla_{\mathbf{X}} \cdot [\eta_f^\alpha \mathbf{W}_f + J \mathbf{F}^{-1} (\mathbf{j}_{f,\text{disp}}^\alpha + \mathbf{j}_{f,\text{diff}}^\alpha)] = J \sum_m \nu_{f(m)}^\alpha r_{(m)}, \quad (58)$$

$$\partial_t (J \phi_s \bar{\rho}_s) |_{\mathbf{X}} = J \sum_m \nu_{s(m)} r_{(m)}. \quad (59)$$

Adding all twelve equations cancels component-relative flux and internal mass production, leaving

$$\partial_t (J \rho) |_{\mathbf{X}} + \nabla_{\mathbf{X}} \cdot \mathbf{W}_f = 0. \quad (60)$$

The same weighting by element counts or specific charges gives the element or charge balances. The variational constraint uses the phase-attached increment $\mathcal{C}_f^\alpha$, while these reference balances follow the solid skeleton.

Define the total nominal stress $\mathbf{P} = J \boldsymbol{\sigma} \mathbf{F}^{-T}$. Quasi-static mechanics is an optional restriction of (54); with inertial terms omitted it reads

$$\mathbf{0} = \nabla_{\mathbf{X}} \cdot \mathbf{P} + J \rho \mathbf{g} - \frac{\sum_{\alpha} \dot{c}_f^\alpha}{\rho_f} \mathbf{F} \mathbf{W}_f. \quad (61)$$

The factor J in the current insertion force cancels against the inverse flux transform, which explains its absence in the last displayed coefficient.

9.1 Residuals and boundary signs

Let scalar tests $\delta\eta_f^\alpha, \delta\phi_s, \delta\tau, \delta\varphi$ and vector test $\delta\mathbf{u}$ vanish on their essential boundaries. Define outward component mass flux $\bar{h}_f^\alpha$, nominal traction $\bar{\mathbf{T}}$, and reference outward electric displacement $\bar{q}_0$. Each fluid component balance is tested independently with arbitrary $\delta\eta_f^\alpha$. The fluid residual is

$$\begin{aligned} 0 = & \int_{\mathcal{B}_0} \delta\eta_f^\alpha [\partial_t (J\phi_f \bar{\rho}_f \eta_f^\alpha) - J \sum_m \nu_{f(m)}^\alpha r_{(m)}] dV \\ & - \int_{\mathcal{B}_0} \nabla_{\mathbf{X}} \delta\eta_f^\alpha \cdot [\eta_f^\alpha \mathbf{W}_f + J\mathbf{F}^{-1}(\mathbf{j}_{f,\text{disp}}^\alpha + \mathbf{j}_{f,\text{diff}}^\alpha)] dV + \int_{\Gamma_h} \delta\eta_f^\alpha \bar{h}_f^\alpha dA. \end{aligned} \quad (62)$$

The solid, transfer and momentum residuals are

$$0 = \int_{\mathcal{B}_0} \delta\phi_s [\partial_t (J\phi_s \bar{\rho}_s) - J \sum_m \nu_{s(m)} r_{(m)}] dV, \quad (63)$$

$$0 = \int_{\mathcal{B}_0} \delta\tau [\partial_t \tau - \tfrac{1}{2} |\mathbf{v}_S|^2] dV, \quad (64)$$

$$0 = \int_{\mathcal{B}_0} \left[\nabla_{\mathbf{X}} \delta\mathbf{u} : \mathbf{P} - \delta\mathbf{u} \cdot J\rho\mathbf{g} + \delta\mathbf{u} \cdot \frac{\sum_\alpha \dot{c}_f^\alpha}{\rho_f} \mathbf{F}\mathbf{W}_f \right] dV - \int_{\Gamma_T} \delta\mathbf{u} \cdot \bar{\mathbf{T}} dA. \quad (65)$$

All time derivatives in these residuals are at fixed $\mathbf{X}$. The local τ equation is determined by initial data at each solid material point. Including inertia adds $\int \delta\mathbf{u} \cdot J \sum_\xi \rho_\xi \mathbf{a}_\xi dV$ to the momentum residual.

The pulled-back Gauss residual is

$$0 = \int_{\mathcal{B}_0} \nabla_{\mathbf{X}} \delta\varphi \cdot (J\mathbf{F}^{-1} \epsilon \mathbf{F}^{-T} \nabla_{\mathbf{X}} \varphi) dV - \int_{\mathcal{B}_0} \delta\varphi J \varrho dV + \int_{\Gamma_q} \delta\varphi \bar{q}_0 dA. \quad (66)$$

The positive boundary term follows because electric displacement is minus the dielectric tensor times the potential gradient. As for mass flux, $\bar{q}_0$ denotes outward electric displacement.

9.2 Closed field system and residual responsibilities

The closed system in this subsection uses quasi-static solid and fluid momentum: the acceleration terms are set to zero in (54) and (57), while material-insertion forces and transfer work are retained. A convenient choice of fields is the displacement $\mathbf{u} = \boldsymbol{\chi} - \mathbf{X}$, $\bar{p}_f$, three solid volume fractions ϕ_A, ϕ_B, ϕ_C , eight independent aqueous mass fractions η_f^α , τ , and φ . Set $\eta_f^N = 1 - \sum_{\alpha \neq N} \eta_f^\alpha$ and $\phi_f = 1 - \sum_s \phi_s$. The three scalar mineral solves determine $\bar{\rho}_s$; the aqueous EOS determines $\bar{\rho}_f$; all stresses, specific potentials, fluxes and rates are constitutive outputs, with the reaction/relative-flux block solved together because the transfer offset depends on $\mathbf{w}_f$ and its resistance depends on reaction rates. Solvability of this nonlinear block must be assessed for the selected parameters and state.

This quasi-static system has 17 scalar fields in three dimensions: three displacement, one pressure, three solid fractions, eight independent mass fractions, one transfer potential and one electric potential. They are supplied by three momentum, three solid mass, nine aqueous mass, one transfer and one Gauss equation. The ninth aqueous mass equation governs aqueous phase storage. Material multipliers can be recovered from $\pi_\xi^\alpha = \rho_\xi^\alpha (\hat{\mu}_\xi^\alpha - \psi_\xi)$; L follows from (19). Distention

is recovered from (9). Initial data must satisfy volume and mass-fraction normalization, positive masses, the mineral and fluid EOS, Gauss compatibility and the specified τ gauge.

Retaining fluid inertia makes (57) an evolution relation for fluid velocity or relative flux. A dynamic formulation therefore includes that vector field, its momentum equation and initial velocity data, together with the inertial form of the mixture momentum balance.

The weak forms define the residual and constitutive responsibilities of the implementation. A mechanics residual implements (65); nine component residuals implement (62); three solid residuals implement (63); local transfer and electrical residuals implement (64) and (66). Constitutive materials provide (28)–(32), (36)–(39), and the coupled (42), (46), (57) block. Differentiation must include the implicit mineral roots, reacted mass weights and phase-conversion resistance. The following section specifies the implemented time integration, finite-element spaces and diffusive facet terms. Active-phase treatment and calibration of the complete reacting constitutive block remain open.

10 Discretization and implementation

The reference balances are implemented in a MOOSE finite-element application [4]. Residual kernels assemble the weak forms on the undisplaced solid reference mesh. Materials supply complete mass storage, nominal stress, reaction sources and pulled-back fluxes. The same mass residuals can therefore test a constitutive reduction and implement the general theory. The implementation provides the balance, mechanics, transfer and Gauss operators; the numerical examples below exercise a restricted chemical material. A calibrated constitutive material coupling all seven reactions to the reacting fluid-flux relation remains to be supplied.

10.1 Conservative time integration and constitutive differentiation

Let M denote either $J\phi_f\bar{\rho}_f\eta_f^\alpha$ or $J\phi_s\bar{\rho}_s$. We difference this entire mass per reference volume. Differencing only the mass fraction would omit changes in pore volume, density and deformation. Backward Euler and variable-step BDF2 use

$$\mathcal{D}_t M^n = \frac{M^n - M^{n-1}}{\Delta t_n}, \quad \text{backward Euler,} \quad (67)$$

$$\mathcal{D}_t M^n = \frac{1}{\Delta t_n} \left[\frac{1 + 2\zeta_n}{1 + \zeta_n} M^n - (1 + \zeta_n) M^{n-1} + \frac{\zeta_n^2}{1 + \zeta_n} M^{n-2} \right], \quad \zeta_n = \frac{\Delta t_n}{\Delta t_{n-1}}. \quad (68)$$

BDF2 starts with one backward-Euler step. Current storage is an automatic-differentiation (AD) material property, while the previous two values are stored histories. The monolithic Newton Jacobian therefore includes the dependence of storage on every coupled field. MOOSE propagates these derivatives through material evaluations using MetaPhysicL [6]. The three mineral-volume roots are obtained with bracket selection followed by Newton updates in AD, following our elastic companion implementation. This differentiates the converged local roots together with the outer residual; it does not freeze mineral compression during the global Newton solve.

10.2 Enriched Galerkin transport

A component balance should conserve mass on each element, including when diffusion crosses a material interface. We use the enriched Galerkin (EG) construction [5], adapting the corresponding

operators in the compositional companion. On the line meshes below, a transported scalar q_h is represented by

$$q_h = q_h^c + q_h^0, \quad q_h^c \in V_h^1 \subset H^1(\mathcal{B}_0), \quad q_h^0 \in V_h^0, \quad (69)$$

where V_h^1 contains continuous piecewise linear functions and V_h^0 contains elementwise constants. On tensor-product meshes the backbone is continuous multilinear. The continuous and elementwise constant tests use the same reconstructed storage and source. Constitutive materials evaluate the total field. Its gradient within each element equals the backbone gradient. The piecewise constant tests expose element mass balances.

For a diffusive reference flux $\mathbf{H} = -\mathbf{D}\nabla_{\mathbf{x}}q_h$, choose one normal $\mathbf{n}$ from the plus element to the minus element, define $[q] = q^+ - q^-$, and let braces denote the arithmetic average. The implemented interior numerical outward flux from the plus element is

$$\hat{h} = \{\mathbf{H}\} \cdot \mathbf{n} + \frac{\sigma}{h_e} \{\mathbf{n} \cdot \mathbf{D}\mathbf{n}\} [q_h], \quad h_e = \min(h^+, h^-), \quad \sigma = 12 \quad (70)$$

for the examples here. The two facet residuals use opposite signs. Symmetric interior penalty also adds the adjoint-consistency term, giving the scalar residual

$$\begin{aligned} R_h(v_h) = & \sum_K \int_K [v_h(\mathcal{D}_t M - S) - \nabla_{\mathbf{x}} v_h \cdot \mathbf{H}] \, dV \\ & + \sum_{e \text{ interior}} \int_e \left([v_h] \hat{h} - \{\mathbf{D}\nabla_{\mathbf{x}} v_h\} \cdot \mathbf{n} [q_h] \right) \, dA + \int_{\Gamma_h} v_h \bar{h} \, dA. \end{aligned} \quad (71)$$

Here S is the reference mass source. For a cellwise constant test, the volume gradient and adjoint term vanish. The remaining terms give that cell's mass balance through storage, source and facet flux. Shared facet fluxes cancel on summation. The code also supplies signed off-diagonal mobility penalty and adjoint terms for coupled diffusion; their manufactured-solution tests are discussed below. The displayed form describes the diffusion block and does not establish an advection-stabilization method for the full reacting system.

The decomposition in (69) admits a constant shift between its two parts. In the closed column we fix one backbone boundary value and leave the enrichment unconstrained. The total field retains a free physical boundary value while this choice fixes its representation. Its initial backbone value matches the prescribed initial field. Zero total outward flux is the physical boundary condition. Fluxless mineral fractions use elementwise constants directly; no enrichment anchor is added to their evolution.

11 Numerical verification with mineral reaction

We test a mineral-forming step of carbonation, mechanism (3), in two settings. A closed uniform reactor discriminates precipitation from dissolution and tests the time discretization against an independent extent equation. A closed column adds chemical-potential-driven diffusion and tests spatial refinement and element conservation against an independent finite-volume calculation. The parameters deliberately isolate the chemical mass and transport operators. They are synthetic and supply numerical verification rather than calibrated silica kinetics or experimental validation of olivine carbonation.

11.1 Exact restricted branch and chemical potential coordinate

Retain positive forsterite, magnesite and silica phases, with $\phi_A = 0.2$, $\phi_B = 0.1$, and initially $\phi_C = 0.05$. The aqueous subsystem retains silicic acid (component 7) and water (component N); the

Implemented object	Governing equation	Responsibility
ADReferenceMass,	(62), (63)	Complete storage and its history difference
ADReferenceMassStorage		
ADReferenceSpeciesBalance	(62), (63)	Reference source and volume flux
ADCarbonationMomentum,	(65), (64)	Nominal stress, insertion force and skeleton transfer rate
ADTransferPotential		
ADReferenceGauss,	(66), (56)	Reference dielectric, charge and flux transformation
ADReferencePullback		
ADLogarithmicMinerals	(28), (32)	AD mineral roots, density, stress and Biot response
ADEnrichedGalerkin*,	(71), (68)	
ADReferenceStorageRate		Paired rows, facet terms and a shared complete storage rate
ADSilicaVerificationChemistry	(38), (42)	Binary aqueous silica reduction described below

Table 2: Residual and material responsibilities. The names identify source objects in the accompanying implementation; their presence does not imply a solved full carbonation network.

other aqueous components and the other six mechanisms are excluded from this particular subsystem. This is a restriction to one reaction, rather than an assertion that the complete nine-component kinetic system preserves zero concentrations. Both retained aqueous masses and all three mineral masses stay positive.

Prescribe $\mathbf{F} = \mathbf{I}$, $J = 1$, $\bar{p}_f = p_0 = 0$, $\theta = 298.15$ K, and initially uniform $\tau = 0$. There is no body force or electric field. Choose the same reference intrinsic density $\rho_* = 1000$ kg/m³ for every phase and $v_\alpha^0 = M_\alpha/\rho_*$ in (35). Then $\bar{v}_f^0 = 1/\rho_*$, $\bar{\rho}_f = \rho_*$, and all mineral roots in (28) equal one. The finite positive elastic moduli can satisfy the inequalities of (22) without affecting this unstressed branch. Thus the total stress is zero for any retained phase fractions. Fluid momentum admits $\mathbf{W}_f = 0$; the solid normalization preserves $\tau = 0$, and the transfer offset vanishes. These prescribed fields satisfy the full mechanical and electrical equations, which are not independently exercised by these examples.

Let $M_7 = 0.096113$, $M_N = 0.018015$, and $M_C = 0.060083$, in kg/mol. Since $M_7 = M_C + 2M_N$, reaction mass transfer also transfers exactly compensating volume for the chosen equal densities. Relative mass diffusion is separately zero sum. Consequently $\phi_f = 0.7 - \phi_C$ remains consistent with constant pressure and total mixture mass, including with spatially varying reaction.

Choose $g_7^0 = g_N^0 = 0$ and $\psi_C^0 = R\theta \ln Q_{\text{eq}}/M_C$, where $Q_{\text{eq}} > 0$ is a dimensionless synthetic equilibrium quotient. Write $x = x_f^7$ and $x_f^N = 1 - x$. Equations (38), (39) and (44) give

$$\mathcal{A}_{(3)} = R\theta \ln \frac{x}{(1-x)^2 Q_{\text{eq}}}, \quad r_{(3)} = k \ln \frac{x}{(1-x)^2 Q_{\text{eq}}}, \quad k = K_{(3)} R. \quad (72)$$

The rate reverses at equilibrium and $r_{(3)} \mathcal{A}_{(3)} \geq 0$. No prescribed reaction history or manufactured chemical source is used in either example.

To keep the diffusive force linear in the transported coordinate while retaining nonlinear storage, define an invertible, scaled chemical-potential difference:

$$q = \frac{M_7}{R\theta} (\hat{\mu}_f^7 - \hat{\mu}_f^N) = \ln x - \frac{M_7}{M_N} \ln(1-x), \quad (73)$$

$$\eta_f^7 = \frac{M_7 x}{M_7 x + M_N(1-x)}, \quad \eta_f^N = 1 - \eta_f^7, \quad \mathbf{j}_f^7 = -\ell \frac{R\theta}{M_7} \nabla \mathbf{x} q, \quad \mathbf{j}_f^N = -\mathbf{j}_f^7. \quad (74)$$

The derivative $dq/dx = 1/x + (M_7/M_N)/(1-x) > 0$ makes this a change of constitutive coordinate on $0 < x < 1$. Newton inversion in AD recovers x and its derivatives. The governing equations still difference $\rho_*\phi_f\eta_f^7$ and $\rho_*\phi_C$, in kg/m^3 , with sources $-M_7r_{(3)}$ and $M_Cr_{(3)}$. Mole amounts below are diagnostics derived from these mass measures, not replacement primary balances. The solvent balance follows by subtracting the silicic-acid balance from the aqueous phase balance; its production is $2M_Nr_{(3)}$.

In both examples $k = 1 \text{ mol}/(\text{m}^3\text{s})$. Set Q_{eq} using $n_7 = 5 \text{ mol}/\text{m}^3$ and $\phi_C = 0.05$, with $n_7 = \rho_*\phi_f\eta_f^7/M_7$, $n_N = (\rho_*\phi_f - M_7n_7)/M_N$, and $x = n_7/(n_7 + n_N)$. Closed-batch equilibria differ slightly from this reference state because the solvent amount changes during reaction. The column uses $\ell = 10^{-6} \text{ kg s}/\text{m}^3$; the batch sets diffusion to zero. These choices define the test rather than measured mineral properties.

11.2 Closed reactor: opposite reaction directions and time refinement

The reactor starts with either $n_7 = 10$ or $n_7 = 1 \text{ mol}/\text{m}^3$. The first state precipitates silica and the second dissolves it. For an independent reference, let $b(t)$ be molar reaction extent per mixture volume, with $b(0) = 0$. Conservation gives

$$n_7 = n_7(0) - b, \quad n_C = n_C(0) + b, \quad n_N = n_N(0) + 2b, \quad \partial_t b = r_{(3)}. \quad (75)$$

Here $n_C = \rho_*\phi_C/M_C$. The independent scalar reference uses DOP853 with relative tolerance 2×10^{-12} and absolute tolerance 10^{-13} , evaluating (72) directly from the conserved mole amounts. The MOOSE calculation instead solves two mass residuals for q and ϕ_C . Its BDF2 histories and the independent reference are compared in Figure 1.

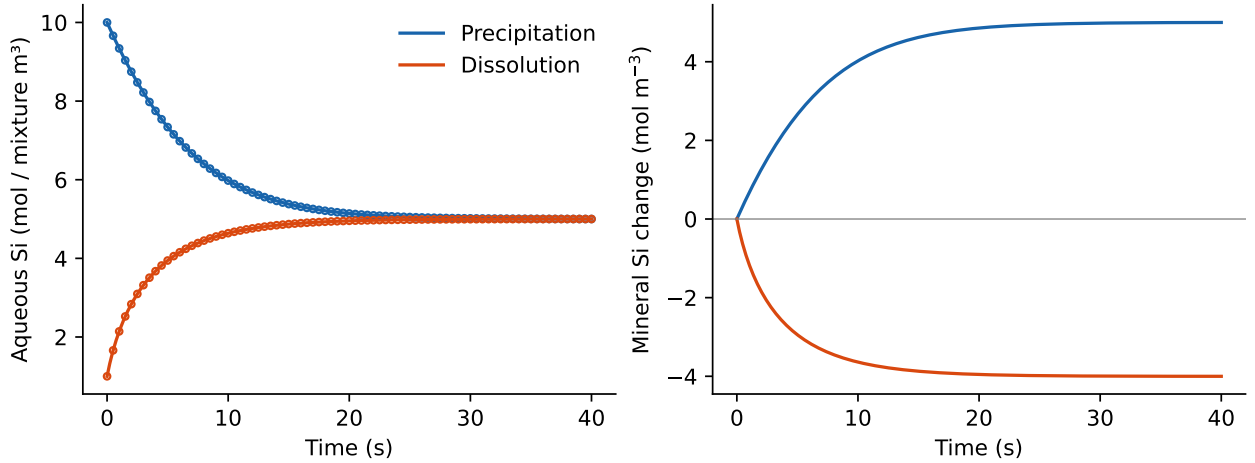


Figure 1: Actual closed-reactor precipitation and dissolution. Lines are MOOSE BDF2 results; open markers in the aqueous panel are independent scalar-extent reference values. The mineral amount increases as dissolved silica decreases, and decreases in the initially undersaturated reactor.

Over 40 s with $\Delta t = 0.0125 \text{ s}$, the precipitating reactor transfers $4.999652 \text{ mol}/\text{m}^3$ of Si into the mineral, while the dissolving reactor transfers $4.000984 \text{ mol}/\text{m}^3$ out of it. Maximum aqueous trajectory errors against the independent reference are 8.07×10^{-6} and $1.69 \times 10^{-4} \text{ mol}/\text{m}^3$, respectively. Both approach their own mass-constrained equilibrium, with final absolute rates below $0.002 \text{ mol}/(\text{m}^3\text{s})$. Si, H and O inventories are conserved to relative error below 10^{-9} , and sampled reaction power is nonnegative.

For a time-accuracy test independent of near-equilibrium flattening, compare the aqueous amount at 5 s on successively halved time steps. Table 3 reports the actual errors. The observed order is computed as $\log_2(E_{\Delta t}/E_{\Delta t/2})$. The finest-pair orders are 1.000 and 0.999 for backward Euler and 2.000 and 1.993 for BDF2. This test exercises the nonlinear mass storage and solution-dependent reaction source together.

Δt (s)	Precipitation error (mol/m ³)		Dissolution error (mol/m ³)	
	Euler	BDF2	Euler	BDF2
0.1	1.13×10^{-2}	2.47×10^{-4}	2.24×10^{-2}	4.49×10^{-4}
0.05	5.66×10^{-3}	6.15×10^{-5}	1.13×10^{-2}	1.20×10^{-4}
0.025	2.83×10^{-3}	1.53×10^{-5}	5.66×10^{-3}	3.11×10^{-5}
0.0125	1.42×10^{-3}	3.83×10^{-6}	2.84×10^{-3}	7.94×10^{-6}
0.00625	7.09×10^{-4}	9.57×10^{-7}	1.42×10^{-3}	2.00×10^{-6}
0.003125	3.55×10^{-4}	2.39×10^{-7}	7.10×10^{-4}	5.04×10^{-7}

Table 3: Closed-reactor time refinement against the independent extent solution at the same final time. Errors are in the aqueous Si amount per mixture volume.

11.3 Closed column: reaction with conservative diffusion

On $0 < X < 1$ m prescribe $n_7(X, 0) = 5 + 4 \cos(\pi X)$ mol/m³, uniform initial mineral fractions, and zero outward mass flux at both ends. The left part initially precipitates silica, while the right part dissolves it. EG transports q , and the elementwise constant silica fraction evolves through its local mass equation. The entire nonlinear storage and reaction block is solved together. The Newton relative and absolute residual tolerances are 10^{-11} , with a direct LU linear solve and backtracking. This solver choice establishes the small verification calculations, not scalable performance for the full field system.

The independent reference uses cell-centered finite volumes, centered potential-difference fluxes, and SciPy BDF method-of-lines integration at relative tolerance 2×10^{-10} and absolute tolerance 10^{-11} . It evolves aqueous and mineral Si diagnostics with opposite reaction terms, calculates solvent mass by closure, and evaluates (72) and (73) directly. It imports no MOOSE residual or constitutive implementation. Exact cell averages initialize its cosine profile. MOOSE instead interpolates the initial potential in the continuous backbone; this introduces an initial projection error that is included in the mesh comparison.

The finite-volume reference uses 512 and 1024 cells; their final aqueous profiles differ by 4.59×10^{-7} mol/m³ in cell-average root-mean-square norm after aggregating fine cells. The EG mesh comparison uses 16, 32, 64 and 128 elements with $\Delta t = 0.005$ s. Reference cell averages are aggregated onto each comparison mesh. Table 4 reports the aqueous and mineral profile errors at 5 s. These comparisons measure agreement with a refined independent discretization. They do not measure error against a closed-form continuum solution.

On the finest mesh the largest local mineral gain is 1.019777 mol/m³ and the largest loss is 1.114963 mol/m³. Global Si, H and O relative inventory drift is below 10^{-9} throughout the column runs. Reconstructing the shared numerical flux in (70) from the sampled backbone and enrichment, and evaluating the BDF2 storage difference and local reaction term on every cell and time step, gives a maximum absolute aqueous mass-balance residual of 3.16×10^{-10} kg/(m³s). This checks local balance as well as global cancellation. Participating phase masses remain positive.

A separate fixed-mesh study on 64 elements uses time steps 0.04, 0.02 and 0.01 s. Successive final-profile differences give a BDF2 self-convergence order of 1.869. This isolates temporal refinement from

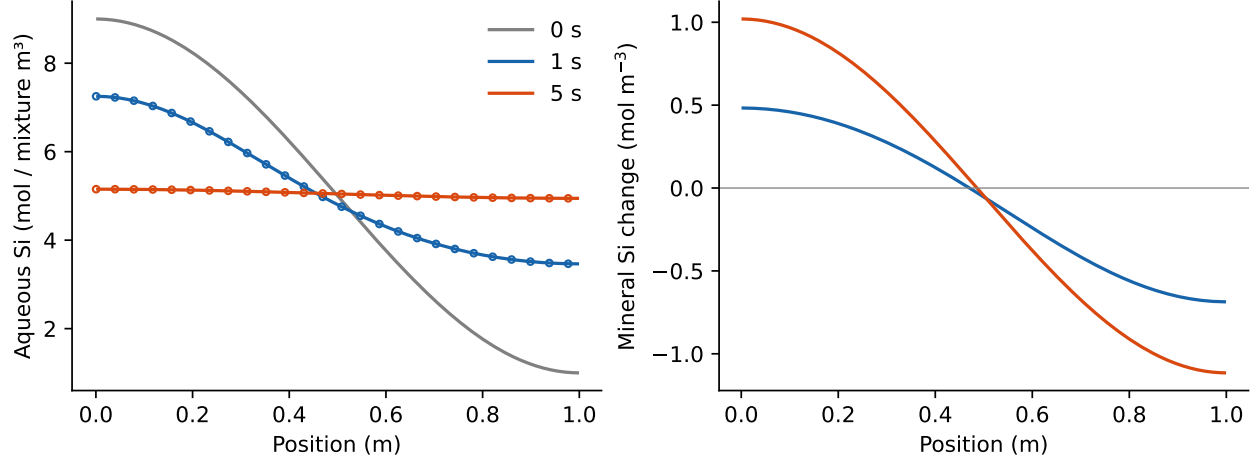


Figure 2: Reaction and counterdiffusion in the closed column. Lines show EG profiles, and open markers in the aqueous panel show the independently refined finite-volume reference. Positive mineral change denotes precipitation and negative change denotes dissolution. The initial aqueous gradient smooths while the locally immobile mineral records the reaction history.

Elements	Aqueous RMS error (mol/m ³)	Mineral RMS error (mol/m ³)	Aqueous order
16	2.88×10^{-3}	5.00×10^{-3}	—
32	7.22×10^{-4}	1.25×10^{-3}	1.994
64	1.81×10^{-4}	3.13×10^{-4}	1.994
128	4.59×10^{-5}	7.80×10^{-5}	1.980

Table 4: Spatial comparison at a fixed small time step. RMS norms compare cell averages of aqueous and mineral Si; the mineral fraction is represented by elementwise constants.

the mesh study. A finite-difference check of the coupled reaction, nonlinear mass and EG Jacobian gives a maximum relative matrix discrepancy of 7.4×10^{-11} . Together with the independent profile comparison, these tests discriminate the material-state derivatives, reaction signs and conservative facet assembly.

11.4 Reproduction and evidence boundaries

The supplement contains the three input decks, all application sources and pinned environment requirements. It also includes raw time histories, selected spatial profiles, compressed solver logs, independent reference data and the scripts that generate the figures and error tables. From repository root, `make moose` builds the application, `make test` reruns the residual suite, and `make verification-examples` reruns the two examples and reference calculations. For an extracted supplement, `environment/README.md` provides direct commands that bypass repository workflow helpers. The executable reports bind source, input, toolchain and data hashes to actual commands and return codes. The framework is pinned at `abafb58b67a6037c6723ffeb19647c84484466da`. Its separately installed dependency is required to rebuild; the supplement does not bundle a binary as a substitute for source reproduction.

The existing residual suite separately checks finite-deformation mechanics and insertion forces, three-mineral roots and Biot response in compression and tension, transfer evolution, dielectric pullback and outward boundary signs. Scalar EG manufactured solutions in one, two and three dimensions and a two-component cross-diffusion problem discriminate spatial accuracy. A time-dependent source with nonlinear mass storage discriminates backward Euler from BDF2. Those tests are distinct from the actual mineral reaction examples. Neither group provides physical validation, an active-phase extension, or a solution of the calibrated full carbonation network. Together they provide a reproducible basis for extending the implementation to those additional cases.

12 Scope of the consistency checks

The companion analytical script checks atom, mass and charge stoichiometry; transfer normalization and offsets; mass-source cancellation; the single-solid elastic limit; constitutive pressure and stress derivatives; the scalar-distention restriction; Biot tangents; reference transformations of virtual work and flux; reacting-flux inversion; the Darcy limit; and nonnegative force-rate products. Its report records inspected file digests, parameters, errors, tolerances and commands. These synthetic checks establish the tested algebraic and constitutive identities; the numerical evidence in Section 11 addresses the stated discrete residuals and restricted reaction solutions.

Quantitative application requires compatible chemical standard potentials, activity models, mineral moduli, permeability, species mobilities and reaction mobilities. Phase disappearance and nucleation require a derived active-phase extension. The present equations describe the elastic, positive-mass branch of the reacting mixture.

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